

Thus on identifying the authorised person, port pin P3.2 goes high, transistor T2 drives into saturation, and relay RL1 energises to open the door for the person. Simultaneously, the LCD shows "access granted" message and port pin P1.7 drives piezobuzzer PZ1 via transistor T1 for aural indication.

If the person is unauthorised, the LCD shows "access denied" and the door doesn't open. LED2 and LED3 show presence of the tag in the RFID reader's electromagnetic field.

To derive the power supply, the 230V, 50Hz AC mains is stepped down by transformer X1 to deliver a

secondary output of 15V, 500 mA. The transformer output is rectified by a full-wave rectifier comprising diodes D1 through D4, filtered by capacitor C1 and regulated by ICs 7812 (IC2) and 7805 (IC3). Capacitor C2 bypasses the ripples present in the regulated supply. LED1 acts as the power indicator and R2 limits the current through LED1.

An actual-size, single-side PCB for RFID-based security system is shown in Fig. 6 and its component layout in Fig. 7. Assemble the circuit on a PCB as it minimises time and assembly errors. Carefully assemble the components and double-check for any overlooked error.

Software

The software for this project is given at the end of this article. It is written in 'C' language and compiled using Keil µVision4 compiler. The finally obtained 'hex' code is burnt into the microcontroller using a suitable programmer. The program is easy to understand.

EFY note. The source code of this project has been included in this month's EFY-CD and is also available on www.efymag.com website.

The complete kit of this project is available with Kits'n'Spares.

The author is founder & CEO of HBeonLabs, Greater Noida

RFID.C

```
#include<reg51.h>
sbit RS=P3^7;
sbit EN=P3^6;
sbit R=P3^2; // for relay
sbit bz=P1^7; // for piezo buzzer
void Rxmag(void);
void lcdinit(void);
void lcdData(unsigned char l);
void lcdcmd(unsigned char k);
void DelayMs(unsigned int count);
void successRx(void);
void unknown(void);
void display(unsigned char s, t);
void welcome(void);
void main()
{
    unsigned char i=0;
    unsigned int j=0;
    unsigned char c[15];
    TMOD=0x20; // Config-
    ure the serial port to 9600 baud rate
    TH1=0x0F;
    SCON=0x50;
    TR1=1;
    R=0;
    lcdinit();
    welcome();
    bz=1;
    while(1)
    {
        back:
        for(i=0;i<15;i++)
        //command to recv data
        {
            c[i]=0xFF;
        }
        while(RI=0);
        for(i=0;i<15;i++)
        //command to recv data
        {
            j=0;
            while(RI=0)
            {
                if(j>=1000)
                goto timeout;
                j++;
            }
            c[i]=SBUF;
            RI=0;
        }
        timeout:
        for(i=0;i<15;i++)
        {
            if(c[i]!='1' && c[i+1]!='E'
            && c[i+2]!='0' && c[i+3]!='0' &&
            c[i+4]!='7' && c[i+5]!='C'
            && c[i+6]!='A' && c[i+7]!='0' &&
            c[i+8]!='3' && c[i+9]!='C')
            // RFID code
            {
                successRx();
                DelayMs(1000);
                R=1;
                bz=0;
                DelayMs(1000);
                R=0;
                bz=1;
                DelayMs(1000);
                lcdinit();
                DelayMs(100);
                welcome();
                goto back;
            }
            unknown();
            DelayMs(2000);
            bz=0;
            DelayMs(2000);
            bz=1;
            DelayMs(1000);
            lcdinit();
            DelayMs(100);
            welcome();
        }
        void successRx()
        {
            unsigned int i=0;
            unsigned char c[]="ACCESS GRANTED ";
            lcdcmd(0x01);
            DelayMs(10);
            lcdcmd(0x80);
            while(c[i]!='\0')
            {
                lcdData(c[i]);
                i++;
            }
            void unknown(void)
            {
                unsigned int i=0;
                unsigned char c[]="ACCESS DENIED";
                lcdcmd(0x01);
                DelayMs(10);
                lcdcmd(0x80);
                DelayMs(10);
                while(c[i]!='\0')
                {
                    lcdData(c[i]);
                    i++;
                }
            }
            //-----
            // Lcd Initialization subroutine
            //-----
            void lcdinit(void)
            {
                lcdcmd(0x38);
                DelayMs(250);
                lcdcmd(0x0E);
                DelayMs(250);
                lcdcmd(0x01);
                DelayMs(250);
                lcdcmd(0x06);
                DelayMs(250);
                lcdcmd(0x80);
            }
            DelayMs(250);
        }
        //-----
        // Lcd data display
        //-----
        void lcdData(unsigned char l)
        {
            P2=1;
            RS=1;
            EN=1;
            DelayMs(1);
            EN=0;
            return;
        }
        //-----
        // Lcd command
        //-----
        void lcdcmd(unsigned char k)
        {
            P2=k;
            RS=0;
            EN=1;
            DelayMs(1);
            EN=0;
            return;
        }
        //-----
        // Delay nS function
        //-----
        void DelayMs(unsigned int count)
        {
            // 1µSec Delay 11.0592 Mhz
            unsigned int i;
            // Keil v7.5a
            while(count) {
                i = 115;
                // 115 exact value
                while(i>0)
                    i--;
                count--;
            }
            void welcome(void)
            {
                unsigned int i=0;
                unsigned char c[]="WELCOME TO RFID";
                unsigned char d[]="SECURITY SYSTEM";
                lcdcmd(0x01);
                DelayMs(10);
                lcdcmd(0x80);
                DelayMs(10);
                while(c[i]!='\0')
                {
                    lcdData(c[i]);
                    i++;
                }
                lcdcmd(0xc0);
                i=0;
                while(d[i]!='\0')
                {
                    lcdData(d[i]);
                    i++;
                }
            }
        }
    }
}
```